

Descriptor	Definition	Example	Format	Cardinality	Source	Comments	ISA Entity	ISA Field	Further description
INVESTIGATION	Investigations are research programs with defined aims. They can exist at various scales (e.g., they could encompass a grant-funded program of work, the various components comprising a peer-reviewed publication, or a single experiment).								
investigationId	A unique identifier of the dataset.	2025SF0002	Unique identifier	1	Plant		investigation	identifier	
investigationTitle	A human-readable string summarizing the investigation.	The behaviour of laboratory mice in KI-TIERWOHL.	Free text (short)	1	Plant FAANG		investigation	title	
investigationDescription	A human-readable text describing the investigation in more detail.	Research on how mice behave in different environments under laboratory conditions.	Free text	0-1	Plant FAANG		investigation	description	
submissionDate	The date on which the dataset was submitted, e.g. with a publication or to a repository.	2025-11-04	Date/Time (ISO 8601, optional time zone)	0-1	Plant		investigation	submissionDate	
publicReleaseDate	The date of the first public release of the dataset presently being described.	2025-11-04	Date/Time (ISO 8601, optional time zone)	0-1	Plant		investigation	publicReleaseDate	
license	A license for the reuse of the data associated with this investigation. The Creative Commons licenses cover most use cases and are recommended.	CC-BY-4.0	Unique identifier	0-1	Plant	Discussion: Can multiple licenses be issued per study within one investigation (e.g. multiple partners)? For submission, one license for the published dataset needs to be the "most" restrictive. If the studies licenses are too different, the study has to be published separately as its own investigation. Idea: using SPDX Link https://spdx.dev/	investigation	comment[license]	
potentialChecklistVersion	A placeholder for further adaption of the checklist.			0	Plant	No MIAPPE available at the moment, only for commentary version- need new name in the future			
associatedPublication	An identifier for a literature publication where the investigation is described. Use of DOIs is recommended.	doi: 10.1371/journal.pone.0071377	DOI	0+	Plant		investigation	publications	
STUDY				0+					
studyId	A unique identifier of the study (e.g. a study could be an experiment).	2025RR0001	Unique identifier	1	Plant		study	identifier	
studyTitle	The title should be a representation of the experimental design.	2025 Laboratory mice Experiments	Free text (short)	1	Plant		study	title	
studyDescription	A human-readable text that is describing the study.	Study on the nutritional intake of mice.	Free text	0-1	Plant		study	description	
studyProject	The referred project, the study, the founder, or a collective that is working on this study.	Project: KI-TIERWOHL	Free text	0-1					
studyStartDate	The date and, if relevant, the time when the experiment started.	2002-04-04; 2006-09-27T10:23:21+00:00	Date/Time (ISO 8601, optional time zone)	1	Plant		study	comment[Study Start Date]	
studyEndDate	The date and, if relevant, the time when the experiment ended.	2025-11-04	Date/Time (ISO 8601, optional time zone)	0-1	Plant	To be discussed: If animals have long-term experiments without end. Potential splitting of data over longer time periods- letting study open?	study	comment[Study End Date]	
contactInst	The name and address of the institution responsible for the study.	IEC - Rostock university medical center; https://ror.org/04dm1cm79	Free text (short)	1	Plant FAANG	Recommendation to use the ROR. If multiple institutions should be named, then they need to model characteristic institutions to add multiple insitutions. For a minimal idea, we stick with one contact institution, e.g. the coordination institution, when dealing with larger projects. Another solution would be the split of studies based on institutions. FAANG also includes for the organization the address, the URI, and the role of the organization.	study	comment[Study Contact Institution]	
locationCountry	The country where the experiment took place, either as a full name or, preferably, as a 2-letter code.	DE	Country name or 2-letter code (ISO 3166)	1	Plant		study	comment[Study Country]	

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siteName	The name of the natural site, experimental facility, laboratory site, external collaborator etc. where the experiment took place.	FBN pig facility; IEC laboratory; farm of Peter	Free text (short)	1	Plant		study	comment[Study Experimental Site]	
locationLatitude	The latitude of the experimental site in degrees, in decimal format.	43.619.264	Degrees in the decimal format (ISO 6709)	0-1 (1 if longitude is provided)	Plant	Discussion: Only relevant when there is interaction with the atmosphere (weather). Recommendation to trim the geolocation coordinate uncertainty.	study	comment[Study Latitude]	
locationLongitude	The longitude of the experimental site in degrees, in decimal format.	3.967.454	Degrees in the decimal format (ISO 6709)	0-1 (1 if latitude is provided)	Plant		study	comment[Study Longitude]	
locationAltitude	Altitude of the experimental site, provided in meters (m).	100 m	Numeric + unit abbreviation	0-1	Plant	Discussion: Remove this here and make it optional for environments where it is necessary.	study	comment[Study Altitude]	
expeDesignDesc	A short description of the animal experimental design, possibly including statistical design. In specific cases, e.g. legacy datasets or data computed from several studies, the experimental design can be "unknown"/"NA", "aggregated/reduced data", or simply "none".	In this experiment, 24 animals were studied after statistical evaluation for identifying correlations between the analyzed factors. The experimental setup included 4 groups, each with 6 animals.	Free text	1	Plant		study	studyDesignDescriptors	
expeDesignType	The type of the experimental design of the study, if available, e.g. the kind of statistical analysis, could be shared.	mesh:D046888	Free text (Ontology term if possible)	0-1	Plant				
obsUnitLevel	The level of the unit used in the study, e.g. on the level of animal groups, single animals or schwarms.	mice XY1 - XY4 in cage 5.	Formatted text (Ontology term / label / Comment)	0-1	Plant	Discussion: Originally this always contains label, ID and source (and comment), which is correlated with an ontology. But you can let the ID and source be "empty" ontologies and just use label and comment to get more information about your study design in there. Ontology in this context could also mean controlled vocabulary or similar things with sources.			
obsUnitDesc	A general description of the observation units in the study and how they were selected, e.g. an animal group of a specific race. If possible, also provide the number of animals in the study.	The observation unit consists of all mice in cage 5 that are siblings and share the same unusual characteristics (e.g. C57BL6/J).	Free text	1	Plant				
husbandryFacility	The type of husbandry in which the study was carried out.	Mixed-housing (indoor and outdoor) - owlEOL:0001697	Free text (Ontology term if possible)	0-1	Plant	Discussion: For livestock animals there is the EOL: http://purl.org/obo/owl/EOL_0001697 . Can it be used even for non-livestock animals, like laboratory mice and primates?	study	studyDesignDescriptors	
husbandryFacDesc	A short description of the facility in which the study was carried out. An extension of the husbandry facility if needed.	The study contains a mixture of barn and outdoor opportunities by free will of choice.	Free text (short)	0-1		Discussion: Is it already covered in siteName	study	studyDesignDescriptors.comments [Description]	
PERSON									
personName	The name of the person (either full name or as used in scientific publications).	Jonny Depp	Name	1	Plant FAANG	Discussion: In FAANG the personName is split into Last and First Name.			
personEmail	The electronic mail address of the person.	bruno@cat.us	E-mail address	0-1	Plant FAANG				
personId	A unique identifier for the data submitter. If that submitter is an individual, ORCID identifiers are recommended.	orcid.org/123-456	Unique identifier	0-1	Plant				

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personRole	The type of contribution of the person to the investigation.	Data submitter; author; corresponding author	Free text (short)	1+	Plant FAANG	Discussion: Do we use specific controlled vocabulary? - Using CreDit, e.g. using the list of roles from Zenodo Potential Recommendation: https://datacite-metadata-schema.readthedocs.io/en/4.5/appendices/appendix-1/contributorType/ In FAANG they used the EFO Ontology (EFO_0001440 - for role).			
personAffiliation	The institution the person belongs to.	IEC - Rostock university medical center; https://ror.org/04dm1cm79 ; https://ror.org/0207ad724	Free text (short)	1+	Plant FAANG				
DATA FILE									
dataFileLink	The link to the data file (or the digital object) in a public database (if available). Otherwise, enter a link to a specific data file inside the published investigation.	https://www.ebi.ac.uk/biostudies/ArrayExpress/studies/E-GEOD-18905?query=mice	URL or file name	1	Plant	Discussion: Is this really just a public link, or is there the possibility to use "internal" links to non-open repositories or even links to files inside the submitted investigation? It could be really just the file path (URI) - but also "internal" links could be used to make the data accessible, even when a prior request to one of the authors has to be made.			
dataFileDesc	The description of the format of the data file. Maybe a standard file format name or a description of the organization in which the data was collected in a tabular file.	FASTA; tab-delimited; column headers headers: 1. A 2.B 3.C; description_file.txt	Free text (short)	0-1	Plant	Could use this with fragments (WC3 Standard ARC) because in the plants they do not use a data standard.			
dataFileVersion	The version of the dataset (the actual data).	./data/filea.csv	Version number	0-1	Plant	https://www.w3.org/TR/annotation-model/#selectors This is reasonable for machine actionability & accessibility			
SOURCE MATERIAL	For biological material, we assume it is the first kind of combination used to form a new animal, e.g. an embryo. When there is no information about the lineage/tree (pedigree), the single animal could also be the start of the biological material. For animals, we have always had a kind of 1:1 connection: mother-father-child. But for plants, there is a 1:n connection - identical seed from the same accession.			0+					Remark: Highly different from plants!
SourceMaterialId	An unique identifier for the source material. It could be an internally generated one or one out of the origin country (Botswana). Possible are also a transport number, the prior living circumstances, etc.	shipping1_of_animal_from_botswana	Unique identifier	1	Own_construction				
SourceMaterialExtId	An external identifier that already exists, e.g. mouse strains. It's possible that the identifier is identical to the SourceMaterialId.	C57BL6/J; wildlings	Unique identifier	0+	Own_construction				
SourceMaterialOrganism	A unique identifier for the organism at the species level. The use of the NCBI taxon ID is recommended.	http://purl.obolibrary.org/obo/NCIT_C45247	URL or File name	1	Own_construction				
SourceMaterialGenus	The genus name for the organism under study, according to the standard scientific nomenclature.	mus, sus	Genus name	0-1	Own_construction				
SourceMaterialSpecies	Species name (formally: specific epithet) for the organism under study, according to standard scientific nomenclature.	mus musculus; sus domesticus	Species name, recommended: binomial form	1	Own_construction				

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SourceMaterialSubspecies	The name of any subtaxa level, including variety, crossing name, etc. The category can be used to store any additional taxonomic identifier. To be filled as a key-value pair list format (the key is the name of the rank/category, and the value is the value of the rank/category).	mus musculus domesticus	Subspecies name	0-1	Own_construction				
SourceMaterialRace	The race of the animals, if available.	Mus strain (C57BL6/J); saddleback pig (http://purl.obolibrary.org/obo/VBO_0001193)	Free text	0-1					
SourcePreprocessing	The description of a process or treatment prior to the start of the study that equally influences the study. It could be provided as free text or as controlled vocabulary.	quarantine time, prior vaccination, fasting, training	Free text (with potential ontology)	0+	Own_construction	The description of conditions prior to the start of the experimental protocol (quarantine, fasting, training, etc.).			
SourceDesc	The description of the material source could also be an internal name of the source.	There are 10 lab mice of the strain C57BL6/J, all female and over 1 month old.	Free text	0-1	Own_construction				
ENVIRONMENT	The environmental parameters were kept constant throughout the study. Remark: If animals change locations, e.g., between barns or within the barn, this should be handled across all animals. So, for each observation unit, there could be multiple protocols for a specific condition.			0-1 per study	Plant				
envParam	The name of the environment parameters, which are constant within the experiment.	temperature; humidity; kind of barn (conventional, eco, free roaming, cage, pen housing, group housing, etc.); light cycles; flooring; cleaning routine; air circulation	Free text (see Appendix I)	1+	Plant				
envParamValue	The value of the environment parameter (defined above), which is constant within the experiment. If possible, use controlled vocabulary.	20°; eco barn	Free text	1 per parameter	Plant				
EXPERIMENTAL FACTOR	The object of a study is to ascertain the impact of one or more factors on the animals. Thus, a factor is, by definition, a condition that varies among observation units, which may be biotic (e.g., pest or disease interactions) or abiotic (e.g., treatments and cultural practices). Depending on the level of the data, an experimental factor can be either an action (e.g., overfeeding) or an indirect action, such as a too-high temperature in the summer when outdoors.			0+ per study; 0+ per observation unit					
expeFactorType	The Name or Acronym of the experimental factor.	Feeding; dietary restrictions; enrichment	Free text (see Appendix II)	1	Plant				
expeFactorDesc	A description of the experimental factor. This includes all relevant treatments and protocols planned for all animals targeted by a given experimental factor.	For the experiment there was one group with high-fat nutrition and one group with low-fat nutrition fed.	Free text	0-1	Plant				
expeFactorValues	A list of possible values for the factor. At least there should be 2 different factor values.	high-fat; low-fat	Free text	2+ per factor	Plant				

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EVENT	An event is a discrete occurrence at a particular time in the experiment (e.g., natural: rain; unnatural: unplanned veterinary treatment). Events may be the realization of factors or parts of factors, or may be confounding to factors. Can be applied at the whole-study level or only to a subset of observation units. Events are mapped to study protocols and could be used to document issues (very freely).			0+ per study/observation unit					
eventType	The short name of the event.	Rearranging animals; vet treating a broken bone; broken equipment (?); sudden death	Free text (short)	1	Plant				
eventDesc	The description of the event, including details such as the applied amount or possibly the duration of the event.	Animals are moved from stable A to B; antibiotics for animal X; GPS tracker stopped working at 2024-11-30	Free text	0-1	Plant				
eventDate	The date and time of the event.	2024-11-30	Date/Time (ISO 8601, optional time zone)	1+	Plant				
OBSERVATION UNIT	The observation units are objects that are subject to instances of observation and measurement. An observation unit comprises one or more animals and/or their environment. (Synonym: Experimental unit)			1+ per study					
obsUnitId	A locally unique identifier, used to identify the observation unit in data files containing the values observed or measured on that unit.	mouse_ABC	Unique identifier	1	Plant	Could this be 1+? For example, you could list just one object, and that unit would be single, or you could list several objects, and the unit would be a group.			
obsUnitType	The type of observation unit in textual form, is usually one of the following: single animal, swarm of animals (unsure number), group of animals (known number).	Mouse_Melissa	Free text	1	Plant				
obsUnitFactorValue	A list of values for each factor applied to the observation unit.	Treatment with drug X	Free text	0+	Plant				
obsUnitSex	The sex of the biological individual(s) could use controlled vocabulary, e.g., using AGROVOC or the potential differentiation of ABCD: <xs:enumeration value="M"/> <xs:enumeration value="F"/> <xs:enumeration value="U"/> <xs:enumeration value="N"/> <xs:enumeration value="X"/> Normal: male, female, hermaphrodite	female	Controlled vocabulary or ontology term	0-1	Own_construction				
obsUnitDate_of_birth	The date of birth of the animal (not applicable if there is a group of animals with different dates of birth).	2024-12-12	Date/Time (ISO 8601, optional time zone)	0-1	Own_construction				
obsUnitAnimal_number	The number of animals per observation unit.	10	Number or NA value	1	Own_construction				

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SAMPLE	A sample is a portion of an observation unit (observation unit = animal) (e.g., tissue, cells, fluids, hairs). A sample must be used when a physical sample needs to be stored and traced. Otherwise, the observations made at the whole living animal-level (non-invasive) should be recorded as animal level observations using the observed variables to characterize the object of the observation (e.g., physiological properties: length of tail, color of the fur).			0+ per observation unit					
sampleId	A unique identifier for the sample.	CEA:BE00034067	Unique identifier	1	Plant FAANG	FAANG uses sample names.			
developmentStage	The life cycle stage of the animal source during which the sample was taken.	Over 1 year; adolescence (between 3-6 months); young and old	Controlled vocabulary	0-1	Plant	Searching if there is a controlled vocabulary. source (NC3Rs): Prenatal development: 5.5 months Infancy: 0-12 months Yearlings and juveniles: 12-36 months Adolescence and sexual maturity: 3-8 years Adulthood Senescence "			
anatomicalEntity	A description of the animal part (e.g., blood, liver) and/or the animal product (e.g., milk) from which the sample was taken.	mouse liver, NCIT:C12392	Free text or ontology if possible	1	Plant				
sampleDesc	Any information not captured by the other sample fields. Could be including quantification, sample treatments, processing and detailed location description.	Left liver lob with a stancer for 1cm x 1cm piece.	Free text	0-1	Plant FAANG				
collectionDate	The date and time when the sample was collected.	2005-08-15T15:52:01+00:00	Date/Time	1	Plant				
externalId	An identifier for the sample in a persistent repository, comprising the name of the repository and the identifier of the observation unit therein. Submission to the EBI Biosamples repository is recommended. URIs are recommended when possible.	Biosamples: SAMEA115580279	Unique identifier	0+	Plant	Discussion: The order of the registration of the material, ENA, and BioSamples. For FAANG it is already defined: first Biosamples IDS and then the FAANG portal.			
OBSERVED VARIABLE	An observed variable describes how a measurement has been made. It typically takes the form of a measured characteristic of the observation unit, associated with the method and unit of measurement. Multiple variables with the same combination of trait, method and scale can be used in association with different animal parts when this distinction is necessary for observations referring to different parts of the same observation unit.			1+ per study		What was measured; how it was obtained; described as a specific phenotypic trait (physiological, behavioral, clinical & method of assessment/unit of measure). Comment: Should be associated with the assay.			
variableId	The code is used to identify the variable in the data file. A variable ID must be unique within a given investigation.	CBC	Unique identifier	1	Plant	CBC = Complete Blood Count (CBC)			
variableName	The name of the variable.	Complete Blood Count	Free text	0-1	Plant	Triglycerides			
traitName	The name of the trait under observation.	lung weight yield	Free text	0-1	Plant				
traitEntity	The entity (part of animal, whole animal) on which the trait has been measured.	lung; social/group level (cage-mates, paired-housing, litter, herd)	Free text	0-1	Plant	Are there ontologies for the following terms? Whole animal (body weight); specific site/tissue (liver, tail, left forelimb); social/group level (cage-mates, paired-housing, litter, herd). For some, there are ontologies, but some are missing, maybe need to adapt an ontology.			

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traitEntityNumber	The accession number of the trait entity in a suitable controlled vocabulary (Plant Ontology).	https://biodivportal.gfbio.org/ontologies/ATOL/ATOL_0005571	Term from Ontology or thesaurus	0-1	Plant	<i>To be discussed: doesn't apply.</i>			
traitCharacteristic	The characteristics that are measured. It can be a morphological characteristic (size, volume, surface), a molecular characteristic (sugar concentration), etc.	size; trigs	Free text	0-1	Plant	trait being quantified; Trigs (mg/dL)			
traitCharacteristicNumber	The number of the trait characteristic in a suitable controlled vocabulary (PATO - the Phenotype And Trait Ontology). A placeholder for further adaptation of the checklist.		Term from Ontology or thesaurus	0-1	Plant	<i>To be discussed: doesn't apply.</i>			
traitNumber	An accession number of the trait in a suitable controlled vocabulary. A placeholder for further adaptation of the checklist.		Term from Ontology or thesaurus	0-1	Plant				
methodName	The name of the method of the observation.	estimation of aggressiveness	Free text	0-1	Plant				
methodNumber	The accession number of the method in a suitable controlled vocabulary. A placeholder for further adaption of the checklist.		Term from Ontology or thesaurus	0-1	Plant	<i>To be discussed: doesnt apply</i>			
methodDesc	A textual description of the method, which may extend a method defined in an external reference with specific parameters, e.g. aggressiveness levels.	aggressiveness levels	Free text	0-1	Plant	Lab animal SOP? Like IACUC, The Guide. Comment: SOPs are now in the study protocol separated.			
methodRef	A URI/DOI of the reference describing the method.	dx.doi.org/10.17504/protocols.io.6qpvr4nr3gmk/v1 https://research.ucdavis.edu/wp-content/uploads/IACUC-31.pdf	URI or DOI	0-1	Plant	IACUC-31; https://research.ucdavis.edu/wp-content/uploads/IACUC-31.pdf			
scaleName	The name of the scale associated with the variable.	cm; qualitative: short, medium, large	Free text	0-1	Plant	Comment: For plants, there is the possibility to differentiate between measurements and estimations (in the method). Therefore, in the scale entity, there should also be a "class" like whether it is a nominal or an ordinal value, but this is missing here (normally it is associated with the crop ontology).			
scaleNumber	The accession number of the scale in a suitable controlled vocabulary. A placeholder for further adaptation of the checklist.		Term from Ontology or thesaurus	0-1	Plant				
timeScale	The name of the scale or unit of time with which observations of this type were recorded in the data file (for time series studies), e.g. daily measurement.	volume milk/day		0-1	Plant				
STANDARD OPERATION PROCEDURE				1+ per study					
SOPname	The name of the procedure.	Imagaging; death of animal; measurement X	Free text	1	Own_construction				
SOPtype	The kind of procedure, e.g. in vitro, in vivo, sample processing protocol, data processing protocol or DNA extraction protocol.	Imaging protocol for mice	term of ontology/controlled vocabulary	0-1	Own_construction	Plants have made their own ontology https://bioregistry.io/registry/dpbo , maybe need our own vocabulary.			
SOPdescription	The description of the protocol.	For the imaging of the whole mouse, the mouse needs to be sedated and put into a CT scanner.	Free text	1	Own_construction				
SOP_URI	An external link to the SOP in human readable format.	protocols.io	URI or DOI	0-1	Own_construction				
SOPversion	The version of the protocol.	2.0	semantic version	0-1	Own_construction				

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SOPparameter	The parameters that need to be defined for each procedure.	body temperature (as an example for a dead animal) Illumination time Resolution Duration in CT Number images Location of camera	Free text	0+	Own_construction				
SOPcomponent	The description of the technical equipment and software used as well.	Camera manufacturer: Nikon Camera model Camera type: infrared	Name and ontology term (if available)	0+	Own_construction				